Spectral–Spatial Joint Sparse NMF for Hyperspectral Unmixing

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***Abstract*— The nonnegative matrix factorization (NMF) combining with spatial–spectral contextual information is an important technique for extracting endmembers and abundances of hyperspectral image (HSI). Most methods constrain unmixing by the local spatial position relationship of pixels or search spectral correlation globally by treating pixels as an independent point in HSI. Unfortunately, they ignore the complex distribution of substance and rich contextual information, which makes them effective in limited cases. In this article, we propose a novel unmixing method via two types of self-similarity to constrain sparse NMF. First, we explore the spatial similarity patch structure of data on the whole image to construct the spatial global self-similarity group between pixels. And according to the regional continuity of the feature distribution, the spectral local self-similarity group of pixels is created inside the superpixel. Then based on the sparse expression of the pixel in the subspace, we sparsely encode the pixels in the same spatial group and spectral group respectively. Finally, the abundance of pixels within each group is forced to be similar to constrain the NMF unmixing framework. Experiments on synthetic and real data fully demonstrate the superiority of our method over other existing methods.**

***Index Terms*— Global spatial structure group, local spectral group, nonnegative matrix factorization (NMF), sparse expression.**

I. Introduction

HYPERSPECTRAL imaging is an emerging technology for acquiring images on hundreds of spectral segments in the same scene simultaneously [1]. A variety of spectra greatly enrich the information of images and have great application potential in many fields [2], such as target recognition [3], image classification [4], medical image analysis [5], and mineral exploration [6]. However, hyperspectral images (HSIs) usually have lower spatial resolution due to the limitations of imagers. In other words, a single pixel may correspond to a larger area of ground containing several kinds of material, which is called mixed pixels [7]. Therefore, to improve the application performance of HSI, hyperspectral unmixing is designed to decompose mixed pixels, which can identify all pure substances (endmembers) in a scene and estimate fractions (abundances) of the endmembers in each pixel [8].

Many previous research studies have revealed the effectiveness of linear mixing models (LMMs) in hyperspectral unmixing [9]. LMM is a popular method to explore the convex geometry of the observed spectra. These methods assume that at least one pure endmember pixel exists in the image, and that these pure pixels are located on the vertices of a simplex. Classic methods include pixel purity index [10], vector component analysis [11], and N-FINDR [12]. In practice, the pure pixels of each endmember in the HSI do not always exist [13]. Therefore, some methods for generating endmembers using the image itself have been proposed, for example, iterative constrained endmembers (ICEs) [14]. However, these methods have high computational complexity, and the unmixing effect is poor on highly mixed images [15]. Treating unmixing as a statistical problem under the Bayesian framework is also a more effective method, but the computational complexity is still prohibitive [16].

In addition, the methods based on sparse regression are also very popular. This kind of methods assumes prior knowledge of the endmembers in the image and pixels as a linear combination of these endmembers, which has guided a lot of work on hyperspectral unmixing [17]–[20]. Unfortunately, these methods rely on the exhaustiveness of a known dictionary containing endmembers in an image. Smaller dictionaries may not contain endmembers, and larger dictionaries will increase endmember consistency, because the high coherence of the dictionaries will lead to poor performance of the sparse regression framework [21].

Recently, hyperspectral unmixing methods are increasingly driven by nonnegative matrix factorization (NMF) and its variants under the linear hyperspectral unmixing model, which can achieve good performance on highly mixed data [22]–[26]. The purpose of NMF is to decompose a nonnegative matrix into two nonnegative factor matrices by learning a parts-based representation of raw data [27]. When applied to hyperspectral unmixing, NMF can estimate both the endmembers and abundances simultaneously without assuming that the image

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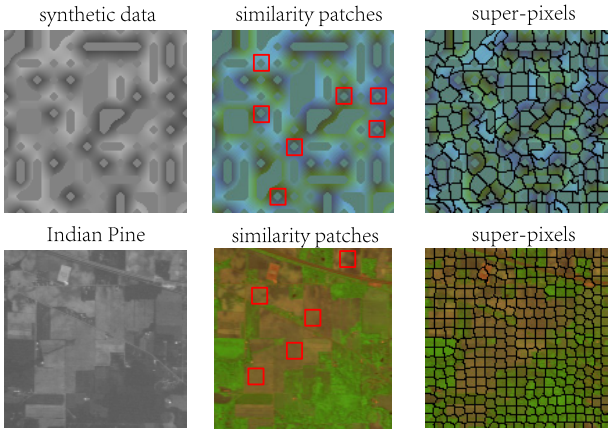


Fig. 1. Example of two types of similarity in HSI. First column: band of synthetic data and Indian Pines, respectively. Second column: similarity patches in the whole image, which contains global spatial information. Third column: result of super-pixel segmentation, which indicates that the pixels of the local region have similar spectral values.

has pure pixels [7]. However, since the nonconvexity of the objective function of NMF, many minima will appear in the result, which leads to the unstable decomposition [28]. In addition, unmixing methods based on NMF usually introduce a noisy factorization and degrade the final performance of unmixing [27]. Therefore, there are more constraints that need to be considered to improve the unmixing performance.

As shown in Fig. 1, there are mainly two types of similarities in HSI. One is the close appearance of patches that exists among the entire image, which can be used to represent the spatial global correlation of data. We call it global spatial similarity (including local and nonlocal spatial information). The other one is the spectral value similarity of adjacent pixels, which can be used to describe the characteristics of the consecutive distribution of the features in the local neighborhood. We call it local spectral similarity. According to the concepts, it can be inferred that these two types of similarity can fully reveal the spatial and spectral correlation of pixels between pixels in the image. In addition, it is noteworthy that pixels with the local/global similarity have similar abundance expressions after unmixing under the NMF framework. If such similarity relation can be integrated into the NMF framework, then the unmixing performance will be more robust and superior.

In this article, we propose a novel hyperspectral unmixing method that utilizes two similarities mentioned above to weight constrained sparse NMF. First, the unmixing is achieved by minimizing the reconstruction error of the sparse NMF. Second, the global patches structure similarity group and the local spectral superpixel group are created by K-means++ clustering and superpixel segmentation, respectively. Third, each pixel is coded using two types of similarity weights by sparse representation on the subspace. Then, the pixel abundance vectors of each global-spatial group and local-spectral group are forced to be similar according to the hyperspectral unmixing principle. Finally, NMF is constrained to make the unmixing result more accurate. It is worth noting that the self-similarity constraint can also significantly suppress the influence of noise on the unmixing in the image, making our results more robust. The complete pipeline of our proposed method includes NMF, abundance representation of global and local self-similarity constraints, and sparse constraint of abundance, as shown in Fig. 2.

In general, the main strengths of the proposed method in this article can be summarized as follows.

1) An NMF framework with self-similarity constraints is proposed, which considers both global spatial information and local spectral information of the data simultaneously.
2) We use K-means++ to explore the global spatial relationship of data through the similarity between patches. And adaptive super-pixel segmentation is used to mine the spectral similarity between pixels in the neighborhood.
3) The pixels in the same global-structure group and the local-spectral group are encoded by the sparse self-expression in the subspace, and the pixels in each group are forced to be similar. Through this global-local perspective, hyperspectral unmixing can make full use of the spatial–spectral context information of the data.

The rest of this article is organized as follows. Section II reviews the related work. In Section III, the sparse NMF is formulated for the problem of unmixing. Our method and optimization are showed in Section IV. Section V evaluates our method on synthetic and real HSIs. The conclusions are given in Section VI.

II. Related Work

Although NMF-based unmixing methods have achieved relatively good results in highly mixed data, it is still a great challenge to obtain high-precision endmember spectra and abundances due to the complexity of terrain scenes and the limitations of techniques. There are many constraints added to the NMF framework to make the results more stable and effective. For example, sparse constraint of abundance has been added to NMF methods to ensure the inherent sparse properties of HSIs [29]–[33]. Among these methods, the $L_{1/2}-$NMF and its extension algorithms are widely used. The core idea of $L_{1/2}-$NMF is to use $L_{1/2}$ regularizer to force abundance vectors to remain sparse. However, the result of $L_{1/2}-$NMF is still unstable due to the different initial settings and noise interference [7]. Pauca *et al.* [34] used a spectral smoothing constraint on the NMF framework to improve the unmixing performance. However, for sparse unmixing, this method requires prior knowledge.

Besides, Miao and Qi [35] proposed a minimum volume constraint with NMF (MVCNMF) for unmixing. Unfortunately, their method requires the dimension reduction, which may lead to the loss of necessary information. Zhuang *et al.* [36] introduced a strategy (NMF-QMV) for automatically selecting regularization parameter, which is used to balance data-fitting term and MV-based regularization. A manifold learning and NMF-based unmixing algorithm is shown by Lu *et al.* [37]. Similarly, Yuan *et al.* [6] showed a substance dependence constraint over NMF to help hyperspectral unmixing, which describes the complicated relationship among pixels in the image.

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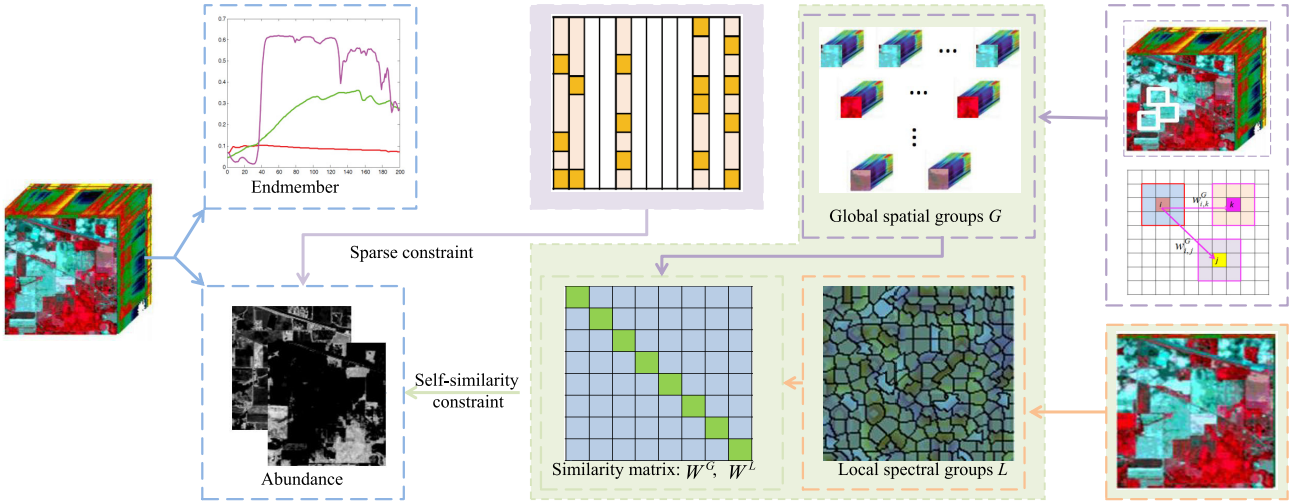


Fig. 2. Schematics of our method. First, we explore the global spatial groups of data on the whole image to construct the spatial similarity matrix W^G. Secondly, according to the regional continuity of the feature distribution, the spectral local similarity matrix W^L is created after the super-pixel segmentation. Finally, the two self-similarity constraints and the sparse constraint are simultaneously applied to the abundance matrix to jointly constrain the NMF unmixing framework.

Abundance separation and smoothness constrained NMF (ASSNMF) [38] is proposed by incorporating abundance separation and smoothness into the NMF algorithm. Its foothold is that the objects in the small area of the HSI are continuous and rarely change suddenly. In [39], a dual graph constraint is added to NMF, which combines the local spatial consistency and the intrinsic manifold structure to constrain the relationship between pixels. Liu *et al.* [40] proposed a weighted NMF (WNMF) algorithm, which incorporated the designed local neighborhood weights to enhance the similarity between pixels in spatial neighborhood. In [41], a neighborhood preserving constrained NMF (NP-NMF) algorithm is proposed, which preserves the local structure of the hyperspectral data on a low-dimensional manifold. SGSNMF takes the superpixel segmentation as preprocessing to generate the spatial groups aiming at constraining the abundance map [42].

Zhang *et al.* [43] also introduced a strategy to preserve local collaboration, which assumes that endmembers generally appear distributed in local spatial regions. A regularization combining nonlocal spatial information is proposed in [44], which explores similar modules and structures in the abundance image. Yang *et al.* [45] combined nonlocal spatial information and group sparsity to constrain NMF framework, in which each spatial group have the same substances. Zare [46] showed the local information proportion (LIP) estimation algorithm, which incorporates a spatial term by estimating based on weighted proportion values of neighboring pixels. To sum up, spatial–spectral joint constraints have shown great potential, which makes the unmixing more precise, since they can combine spatial and spectral information simultaneously.

Our proposed method based on global-structure and local-spectral is related to spatial–spectral information enhancement strategy and $L_{1/2}$ sparse strategies in data. Nevertheless, our approach has major differences from the existing ones. We minimize the NMF reconstruction error by the following characteristics: 1) global spatial groups and local spectral groups are explored in the input HSI to characterize data-rich context information and 2) based on the self-expression of data in the subspace, we combine the sparse coding with the spatial–spectral information to make abundance vectors in each spatial group and spectral group similar, which can improve the robust performance of unmixing.

III. Problem Formulation

Given hyperspectral data $Y = [y_1, y_2, \ldots, y_N] \in \mathbb{R}^{I_3 \times N}$, where I_3 denotes the number of bands and N is the number of pixels, our goal is to estimate all the endmembers contained in the image and the fractions of these endmembers in each pixel. According to the LMM of hyperspectral unmixing [1], the pixel y_n in the image can be expressed as

$$y_n = \sum_{r=1}^{R} m_r a_{r,n} + e_n \quad \text{s.t.,} \quad a_{r,n} \geq 0, \quad \sum_{r=1}^{R} a_{r,n} = 1 \tag{1}$$

where $a_{r,n}$ is the abundance coefficient contributed by endmember m_r, R represents the number of endmembers, and $e_n \in \mathbb{R}^{I_3 \times 1}$ is the additive noise. And two constraint terms: abundance nonnegative constraint (ANC) and abundance sum-to-one constrain (ASC), their purpose is to ensure that (1) is physically interpretable. ANC means that each abundance coefficient should be nonnegative and ASC indicates that the sum of all abundance coefficients in a single pixel should be one.

For the entire image, (1) is written in a matrix form as

$$Y = \mathrm{MA} + E \quad \text{s.t.} \; a_{r,n} \geq 0, \quad \sum_{r=1}^{R} a_{r,n} = 1 \tag{2}$$

where $M = [m_1, m_2, \ldots, m_R] \in R^{I_3 \times R}$ represents the endmember matrix, $A = [a_1, a_2, \ldots, a_N] \in \mathbb{R}^{R \times N}$ is the abundance matrix, each column of which is the fraction vector of all endmembers in one pixel, and $E \in \mathbb{R}^{I_3 \times N}$ denotes the noise matrix.

Among the many existing methods, NMF is widely used as a popular solution to solve (2) since it can obtain the part-based

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representation of hyperspectral data [47]. Given an observation data Y, NMF aims at finding two nonnegative matrices M and A, one representing the base of the data (endmember) and the other denoting the expression coefficient (abundance) of the data under the base [6]. Therefore, (2) can become an optimization problem as follows[1]:

$$\min_{M,A} \ ||Y - \mathrm{MA}||_F^2 \tag{3}$$

where $\|\cdot\|_F$ represents the Frobenius norm. For problem (3), there are many ways to optimize it [48]. Among them, because of simplicity and ease of solution, the multiplication updating rules (MRUs) in [49] is widely used

$$\begin{aligned} M &= M. * \left(\mathrm{YA}^{\mathrm{T}}\right)./\left(\mathrm{MAA}^{\mathrm{T}}\right) \\ A &= A. * \left(M^{\mathrm{T}}Y\right)./\left(M^{\mathrm{T}}\mathrm{MA}\right) \end{aligned} \tag{4}$$

where $(\cdot)^{\mathrm{T}}$ denotes the transpose operator of matrix, $.*$ and $./$ are the elementwise multiplication and division, respectively. However, problem (3) is a nonconvex function with respect to M and A together. Even if iterating under MURs, there will be many local minimum values, which will result in unstable unmixing results and is susceptible to noise [28]. Therefore, many researchers later devote themselves to finding more constraints to add to (3) to get a more accurate and feasible solution.

In recent years, sparsity-constrained NMF has received more and more attention, since sparseness of abundance is a very important property in HSI [7]. Especially in $L_{1/2}$-NMF, $L_{1/2}$ regularizer is added to the NMF to force each abundance vector to be sparse. The objective function of $L_{1/2}$-NMF can be written as

$$\min \ ||Y - \mathrm{MA}||_F^2 + \lambda ||A||_{\frac{1}{2}} \tag{5}$$

where $||A||_{(1/2)} = \sum_{i=1}^{r}\sum_{j=1}^{n} A_{i,j}^{(1/2)}$ is the $L_{1/2}$ regularizer, and λ denotes the regularization parameter. However, $L_{1/2}$-NMF still has some of the same problems as the standard NMF. For instance, different initialization matrices often result in unstable decomposition results. And it is very susceptible to noise interference. Therefore, our goal is to add more suitable regularization terms to NMF model so that we can get a more stable and accurate result for unmixing.

IV. Proposed Method

As shown in Fig. 2, it is the unmixing flowchart of our proposed method. Our main contribution is to add a self-similarity constraint on the hyperspectral unmixing with combining spatial and spectral information for pixels. Defining $\Lambda(M, A) = ||Y - \mathrm{MA}||_F^2$, our objective function can be expressed as

$$\Gamma = \Lambda(M, A) + \lambda_1 ||A||_{\frac{1}{2}} + \lambda_2 \Omega(Y) \tag{6}$$

where the first item denotes the standard form of NMF, the second item is the sparse constraint on the abundance matrix A, the last one indicates the self-similarity regularization for data Y, λ_1 and λ_2 are the weighted parameters of the sparse and self-similarity constraints, respectively. Equation (6) will be optimized in the following three steps: 1) extracting the spatial and spectral self-similarity groups of the pixels form HSI, separately; 2) sparse coding of pixels in the same group; and 3) convex optimization of sparse NMF with self-similarity constraints. We will describe these operations in detail in the following.

A. Self-Similarity Constraints

In HSI, the complex spatial distribution and rich spectral information of objects should be considered. Therefore, we design $\Omega(Y)$ in (6) as a regularization with two types of self-similarity.

1) Spatial Self-Similarity Analysis: First, the spatial similarity analysis will be performed on the whole image. We all know that for a local window $W_1 \times W_2$ on an image, there will be one or more windows of the same size on the entire image that are similar to what they contain. As shown in Fig. 1, this similarity relationship is shown. It is noteworthy that on HSI, there are not only similarities between adjacent patches, but also similarities between nonlocal patches. Thus, we refer to this similarity as the spatial global self-similarity of pixels.

Specifically, the initial form of HSI Y: $X \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, which denotes that data have I_3 bands, and the image size on each band is $I_1 \times I_2$. Considering the speed and complexity of the algorithm, and combining with [50], we use a window size of 3×3 with a step size 1 to segment data X into cubes. After image partition, a new data $V \in R^{(3^2 * I_3) \times (I_1 * I_2)}$ is created by reshaping each of cube into a 1-D vector. It is worth noting that each column vector in V is formed by eight neighborhood reconstructions around each pixel of the original data X. The specific details are shown in Fig. 2. By the above method, the horizontal and the vertical spatial information are all retained in the data V.

In [51], a local window search is used to explore nonlocal spatial information of the data. Unfortunately, the global similarity cannot be obtained due to the limitations of the search area. In addition, the Euclidean distance brings a lot of complexity when calculating the similarity point by point in all samples. Therefore, we use classical clustering method to describe the similarities between windows of the raw data X (points in V) in this article.

Since K-means algorithm are sensitive to the selection of the initial cluster center, we use the K-means++ method [52] to obtain clusters $\chi = \{G_1, G_2, \ldots, G_P\}$, $G_p \in \mathbb{R}^{(3^2 \times I_3) \times g^n}$, $p = 1, 2, \ldots, P$, where P denotes the number of clusters and g^n is the number of global similar 1-D vectors in the pth cluster (each g^n may have different length). With this method, we explore the global spatial similarity of image X. In short, the similarity of samples in V is the nonlocal patch similarity of X. And our goal is to determine the interdependencies relationship in each group.

2) Spectral Self-Similarity Analysis: Assuming that the image pixels of a local region contain similar features and spectral values, the vectors of the multidimensional HSI pixels are similar in that local region. As shown in Fig. 1, this similarity relationship is shown. In this article, we use superpixel

[1] M and A have the same constraints as in (2). In this article, unless otherwise specified, all matrices and vectors satisfy nonnegative constraints.

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to describe the spectral similarity since it is more adaptive in shape and size than fixed square.

To effectively apply superpixel segmentation on HSI and increase the algorithm's operating rate, we first perform the sparse principal component analysis (PCA) to obtain its first three principal components and combine them into one RGB image. And the RGB image is prepared for the SLIC superpixel segmentation algorithm [53]. After obtaining the superpixel, we map the superpixel index $o = \{L_1, L_2, \ldots, L_Q\}$ back to the original hyperspectral data X, where Q is the number of superpixels, and L_q is also a vector containing the pixel indices of the qth superpixel, $q = 1, 2, \ldots, Q$. Within each superpixel group, all pixels share similar spectral features.

According to [54], each pixel can be approximated by the linear representation of all pixels in the same subspace in which it resides. Therefore, in this article, we approximate each pixel using a spatial self-similarity group and a spectral self-similarity group, respectively. To be specific, for any pixel $\mathbf{y}_n$ in the image, it can be expressed by spatial self-similarity and spectral self-similarity, respectively

$$\begin{aligned} y_n &= \sum_{i \in G_p} w^G_{n,i} y_i, \quad n \in G_p, \ p = 1, 2, \ldots, P \\ y_n &= \sum_{j \in L_q} w^L_{n,j} y_j, \quad n \in L_q, \ q = 1, 2, \ldots, Q \end{aligned} \tag{7}$$

here, $w^G_{n,i}$ represents the weight between y_n and the pixel y_i in the global spatial group, $w^L_{n,j}$ denotes the weight between y_n and the pixel y_j in the local spectral group. Furthermore, for y_n in the image, its two types of similarities can be written as

$$y_n = \mu \sum_{i \in G_p} w^G_{n,i} y_i + (1-\mu) \sum_{j \in L_q} w^L_{n,j} y_j \quad \text{s.t. } n \in G_p \wedge n \in L_q \tag{8}$$

where μ $(0 < \mu < 1)$ is a parameter used to adjust the contribution of spatial and spectral self-similarity.

More specifically, $w^G_{n,i}$ $(0 < w^G_{n,i} < 1)$ describes the similarity between two patches centered around y_n and y_i in the HSI. In this article, we can use the Euclidean distance to express as

$$w^G_{i,n} = \frac{1}{H^G} \exp^{\frac{-||v_i - v_n||^2}{\sigma^G}}. \tag{9}$$

Here, H^G is a regularization term defined as $H^G = \sum_{i \in g_p} \exp^{(-||\mathbf{v}_i - \mathbf{v}_n||^2))/\sigma^G)}$ in order to ensure $\sum_i w^G_{i,n} = 1$, and σ^G is a smoothing kernel [2]. since each patch is made up of all the pixels in the 3×3 window, each v is a vector of $3 \times 3 \times I_3$. Similarly, the definition of the local spectral weight $w^L_{n,i}$ is similar to the above. The only difference is that v_n and v_i at this time are each pixel in the hyperspectral superpixels.

It is worth noting that $w^G_{n,i}$ and $w^L_{n,i}$ are both zero when y_n, y_i, y_j are not in the same space group or spectral superpixel. Therefore, we extend the weights between pixels to the entire image, which can be represented as

$$Y = \mu \mathrm{YW}^G + (1-\mu)\mathrm{YW}^L \tag{10}$$

where μ $(0 < \mu < 1)$ is a parameter used to adjust the contribution of spatial and spectral self-similarity.

According to the objective function (5), we can transfer the global and local self-similarity between pixels in the image to the abundance matrix. Because for the same endmember matrix M, the pixel similarity relationship will inevitably have a similar abundance relationship. That is, similar pixels share the same abundance information. Finally, the abundance regularization term with two self-similarity constraints in unmixing can be described as

$$\Omega(A) = ||A - \mu \mathrm{AW}^G - (1-\mu)\mathrm{AW}^L||^2_F. \tag{11}$$

Note that the weights that do not belong to the same relationship group when calculating weights are directly set to 0, so W^G and W^L are sparse. Finally, the objective function of our proposed method can be described as

$$\Gamma = \Lambda(M, A) + \lambda_1 ||A||_{\frac{1}{2}} + \lambda_2 \Omega(A) \quad \text{s.t. } M \geq 0, \quad A \geq 0, \ \sum_{r=1}^{R} a_{r,n} = 1. \tag{12}$$

B. Convex Optimization

The objective function (12) is nonconvex with respect to M and A simultaneously. Therefore, we deduce the update rules using Lagrange multiplier method to find the optimal solution through an alternating iteration process. For convenience, we set $H = I - \mu W^G - (1-\mu)W^L$. Thus, (12) can be rewritten as

$$\Gamma = \mathrm{Tr}(Y - \mathrm{MA})(Y - \mathrm{MA})^{\mathrm{T}} + \lambda_1 ||A||_{\frac{1}{2}} + \lambda_2 \mathrm{Tr}\left(\mathrm{AHH}^{\mathrm{T}} A^{\mathrm{T}}\right). \tag{13}$$

Considering that (12) has two nonnegative constraints: $M \geq 0$ and $A \geq 0$, its corresponding Lagrange multiplier is defined as: $\Psi \geq 0$ and $\Phi \geq 0$, respectively. Therefore, the Lagrange function Ξ can be expressed as

$$\Xi = \mathrm{Tr}\left(\mathrm{YY}^{\mathrm{T}}\right) - 2\mathrm{Tr}\left(\mathrm{YA}^{\mathrm{T}} M^{\mathrm{T}}\right) + \mathrm{Tr}\left(\mathrm{MAA}^{\mathrm{T}} M^{\mathrm{T}}\right) + \lambda_1 ||A||_{\frac{1}{2}} + \lambda_2 \mathrm{Tr}\left(\mathrm{AHH}^{\mathrm{T}} A\right) + \mathrm{Tr}\left(\Psi M^{\mathrm{T}}\right) + \mathrm{Tr}\left(\Phi A^{\mathrm{T}}\right). \tag{14}$$

Then, taking the partial derivative of M and A, respectively, the results are

$$\frac{\partial \Xi}{\partial M} = -2\mathrm{YA}^{\mathrm{T}} + 2\mathrm{MAA}^{\mathrm{T}} + \Psi \tag{15}$$

$$\frac{\partial \Xi}{\partial A} = -2M^{\mathrm{T}} Y + 2M^{\mathrm{T}}\mathrm{MA} + \frac{\lambda_1}{2} A^{-\frac{1}{2}} + 2\lambda_2 \mathrm{AHH}^{\mathrm{T}} + \Phi. \tag{16}$$

According to the Karush–Kuhn–Tucker (KKT) conditions, $\Psi M = 0$ and $\Phi A = 0$. Then we can get

$$M \leftarrow M. * \left(\mathrm{YA}^{\mathrm{T}}\right). \Big/ \left(\mathrm{MAA}^{\mathrm{T}}\right) \tag{17}$$

$$A \leftarrow A. * \left(M^{\mathrm{T}} Y\right). \Big/ \left(M^{\mathrm{T}}\mathrm{MA} + 0.25\lambda_1 A^{-\frac{1}{2}} + \lambda_2 \mathrm{AHH}^{\mathrm{T}}\right). \tag{18}$$

C. Implementation Issues

Here, we add some details to the proposed unmixing algorithm. The first issue is how to ensure that the algorithm persists to meet ASC and ANC during the optimization process. For the update rules (17) and (18), assuming that the initialized matrices M, A are all nonnegative, then it is clear that the ANC

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constraints can be strictly guaranteed. For the ASC constraint, we adopt the approach in [55]. When using (18) to update the abundance matrix, data Y and the endmember M are replaced by $\tilde{Y}$ and $\tilde{M}$, respectively, as defined in the following:

$$\tilde{Y} = \begin{bmatrix} Y \\ \delta 1_N^{\mathrm{T}} \end{bmatrix}$$
$$\tilde{M} = \begin{bmatrix} M \\ \delta 1_R^{\mathrm{T}} \end{bmatrix}. \tag{19}$$

Among them, δ is used to adjust the accuracy of ASC. It is noteworthy that the larger the value of δ, the higher the accuracy of the result, but it also causes the convergence speed of the algorithm to decrease. Therefore, considering the accuracy and speed of the algorithm at the same time, we set the value of δ to 15 in this article.

The second issue is how to stop the algorithm iteration. In general, there are two ways to stop the iteration of the algorithm [6]. First, an error tolerance value can be set. When the error is below this threshold for more than ten consecutive times, the iteration is terminated. The second is to set a maximum number of iterations. In this article, we use the second method and make the number of iterations 3000. It has been experimentally verified that 3000 is sufficient to satisfy the convergence performance of the proposed algorithm.

The third issue is how to choose parameters. First, for an unknown image, if the number of endmembers R is unknown, corresponding estimation method can be used, for instance, HySime [56]. Second, the superpixel segmentation effect can be directly observed from the naked eye. After combining [57], and considering the performance and computational burden of the algorithm, we set the number of clusters to be determined by the number of superpixels in each set of experiments. Specifically, $P = Q = 3349$ in the cuprite data and $P = Q = 2679$ in the India data.

The last issue is how to initialize related matrices. As we all know, NMF-based methods are easily affected by the initialization of two factor matrices. Therefore, we first use VCA-based method to extract endmembers, and FCLS to solve the abundance matrix in this article. Finally, use the results as the initial value of all algorithms to ensure the comparability and stability of each method.

The entire process of the proposed method for unmixing can be summarized in Algorithm 1.

Algorithm 1 S^2-NMF for Hyperspectral Unmixing

Input:

The hyperspectral data $\mathrm{Y} \in \mathbb{R}^{\mathrm{M}\times\mathrm{N}}$, the number of endmembers R, the number of $\mu > 0$, and the parameters $\lambda_1 > 0$, $\lambda_2 > 0$.

Output:

Endmembers matrix M, abundance matrix A.

1: Build global spatial self-similarity groups $\chi = \{G_1, G_2, \ldots, G_P\}$ and local spectral self-similarity groups $o = \{L_1, L_2, \ldots, L_Q\}$;

2: Calculate the weight matrices W^G and W^L using Eq. (9);

3: Solve Eq. (12) via alternating minimization algorithm:

Initialize M and A by VCA-FCLS method.

repeat

(1) Update M by applying the updating rules in Eq. (17);

(2) Update A by applying the updating rules in Eq. (18);

until Converges.

V. Experiments

In this section, we conduct a series of experiments on synthetic data and real data to verify the effectiveness and performance of our algorithm, compared with five existing methods: I-NLSU [58], $L_{1/2}$-NMF, CSNMF [59], SGSNMF, and NMF-QMV. Among these algorithms, CSNMF are SGSNMF are both based on NMF and consider the spatial information, while I-NLSU is a sparse regression method with nonlocal spatial constraint.

Experiments are taken using two popular criteria to evaluate the unmixing results: spectral angular distance (SAD) used to measure the accuracy of estimated endmembers and root mean square error (RMSE) used to evaluate the accuracy of abundance matrix. Specifically, for the rth endmember in the image, SAD between the estimated endmember m_r by the algorithm and the corresponding reference endmember $\hat{m}_r$ is defined as

$$\mathrm{SAD}_r = \cos^{-1}\left(\frac{\hat{m}_r^{\mathrm{T}} m_r}{|\hat{m}_r| \cdot |m_r|}\right) \tag{20}$$

where $|\cdot|$ denotes the norm operator for vectors. And the estimated abundance a_r and the reference abundance matrix $\hat{a}_r$ can be written as

$$\mathrm{RMSE}_r = \sqrt{\frac{1}{n}|a_r - \hat{a}_r|^2}. \tag{21}$$

In the following experiments, we use a similar method in [28]: Monte Carlo simulations, which demonstrates the effectiveness and stability of our proposed algorithm. Each experimental result is subjected to 20 Monte Carlo runs to obtain the mean and standard deviation as the final results.

A. Experiment on Synthetic Data

In this section, multiple experiments are performed on synthetic data. To generate synthetic data, we randomly selected R spectra from the USGS spectral library [60]. The generation methods and steps are similar with [37]. First, an image of size $d^2 \times d^2$ is divided into regions of $d \times d$. And for each region, one endmember spectrum is randomly selected to fill it up. Second, a low-pass filter of $(d-1)\times(d-1)$ is used to generate the mixed image. Thereafter, to make the synthesized data highly mixed, we replace the pixels with abundance greater than θ with the average mixed pixels of all endmembers. Finally, we add zero-mean Gaussian noise to the synthesized data, making it closer to real data. The signal to noise ratio (SNR) of the image is defined here as

$$\mathrm{SNR} = 10\log_{10}\frac{E\left[y^{\mathrm{T}} y\right]}{E\left[e^{\mathrm{T}} e\right]} \tag{22}$$

where y and e are the observation and noise of a pixel, respectively, $E[\cdot]$ is the expectation operator.

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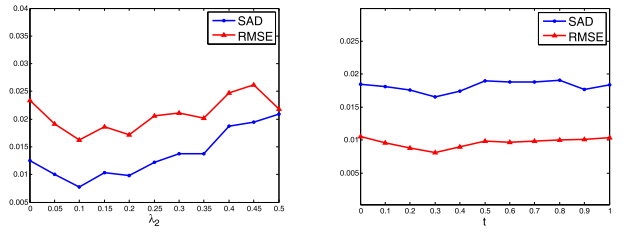


Fig. 3. Performance of S^2-NMF with respect to parameter λ_2 and t. (a) λ_2. (b) t.

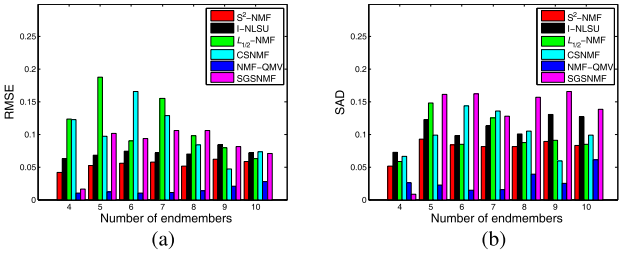


Fig. 4. Comparison of S^2-NMF with different numbers of endmembers in terms of (a) RMSE and (b) SAD.

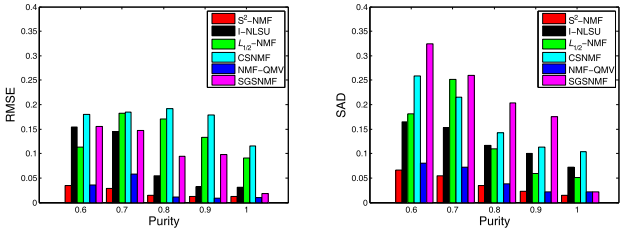


Fig. 5. Comparison of S^2-NMF with different mixing degree in terms of (a) RMSE and (b) SAD.

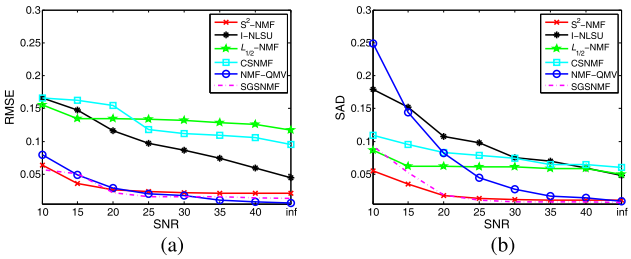


Fig. 6. Comparison of S^2-NMF with different noise levels in terms of (a) RMSE and (b) SAD.

1) Experiment 1: Parameter Setting: In this experiment, when $R = 4$, $d = 9$, $\theta = 0.8$, and SNR $= 30$ dB, we determine the parameters λ_1, λ_2 and t in the proposed algorithm. The experiment uses the control variable method and uses SAD and RMSE as evaluation indicators to determine the best parameters. First, λ_1 is used to regulate the sparsity of abundance, so here we combine the parameter settings in [6] and [60], to set $\lambda_1 = 0.11$. Then, parameter t used to balance the spatial information and the spectral information is set to 0.5, and $\lambda_2 = \{0, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5\}$. Then the unmixing results is shown in Fig. 3(a).

As can be seen from Fig. 3(a), the values of the two measurement matrices are the smallest, when $\lambda_2 = 0.1$. And it shows that the algorithm performs best at this time. Therefore, fixed $\lambda_2 = 0.1$, we set $t = \{0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7$ $0.8, 0.9, 1\}$, respectively. Fig. 3(b) shows the performance of the S^2-NMF algorithm as it changes with t. We can conclude that when $t = 0.3$, the curve gets the minimum, so it can be determined that the value of the adjust parameter t of the spatial–spectral self-similarity constraint is 0.3. In all the experiments that follow, we fix the parameters: $\lambda_1 = 0.11$, $\lambda_2 = 0.1$, and $t = 0.3$.

2) Experiment 2: Robustness to Different Number of Endmembers: This experiment is used to explore the generalization ability of our proposed S^2-NMF for different numbers of endmembers present in images at the same scale. We set SNR $= 30$ dB, $d = 9$, and $\theta = 0.8$. And the number of endmembers takes the value: $R = 4, 5, 6, 7, 8, 9, 10$, thereby synthesizing seven kinds of data containing the number of different endmembers. For each type of data, six different methods are used for unmixing. As a result, the unmixing performance of each algorithm is shown in Fig. 4. It can be seen that the proposed algorithm S^2-NMF maintains a high level of stability and superiority. Besides, the performance of NMF-QMV is also very effective.

3) Experiment 3: Generalization to Different Mixing Degree: For HSIs of varying degrees of mixing, the unmixing algorithm exhibits different performance. In general, the more complex the data, the harder it is to unmix. We design this experiment to evaluate the unmixing performance of our algorithm for images of different mixing degrees. When the data are synthesized, we set SNR $= 30$ dB, $d = 9$, and $R = 4$. And the parameter θ indicating the degree of mixing is increased from 0.6 to 1. The larger the θ value, the lower the mixing of the pixels. When θ is 1, it indicates that there are pure pixels in the image. For the synthetic data under different mixing degrees, the unmixing performance of the six algorithms is shown in Fig. 5. From Fig. 5, we can see that the unmixing performance of the six algorithms is generally improving. It is noteworthy that the proposed algorithm has an ideal unmixing effect even in the case of high image mixing degree.

4) Experiment 4: Generalization of Noise: We use this experiment to illustrate the robustness of the proposed S^2-NMF algorithm under varying degrees of noise interference. When synthesizing data, select $R = 4$, $\theta = 1$, $d = 9$. First, let the value of SNR be infinite, assuming no noise interference in the image. The SNR is then increased from 10 to 40 dB at intervals of 5. After obtaining eight sets of data, they are unmixed by six algorithms, respectively. As shown in Fig. 6, our algorithm performs best, maintaining good unmixing results even at low SNR. Although $L_{1/2}$-NMF, CSNMF are relatively stable, their errors are large, and the unmixing performance is relatively inferior. The SGSNMF algorithm is robust to noise. Although NMF-QMV performs well on RMSE, the algorithm is greatly affected by noise for SAD.

5) Experiment 5: Different Image Size Analysis: This experiment is used to evaluate the unmixing performance of our proposed S^2-NMF algorithm under different size image. The synthesized data are generated by fixing $R = 4$, $\theta = 1$, SNR $= 30$ dB, and the image size is 20×20, 30×30, 42×42, 56×56, 72×72, 90×90. As shown in Fig. 7,

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TABLE I
RUNNING TIME WITH RESPECT TO IMAGE SIZE

Image size	S^2-NMF	I-NLSU	$L_{1/2}$-NMF	CSNMF	NMF-QMV	SGSNMF
20×20	7.9402	3.1252	4.7515	40.7748	7.3271	6.5178
30×30	13.1320	5.2346	8.1153	93.1804	9.6294	8.8624
42×42	30.4807	17.8703	14.1374	174.4576	16.0422	22.9924
56×56	104.5999	89.2530	24.5440	316.2888	26.1381	35.2693
72×72	319.8287	257.9214	34.4722	447.2634	35.9729	57.5386
90×90	1005.8464	789.9162	55.1806	696.0928	63.1148	57.6267

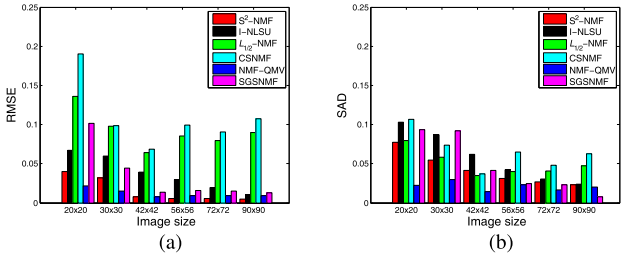


Fig. 7. Comparison of S^2-NMF with different image size in terms of (a) RMSE and (b) SAD.

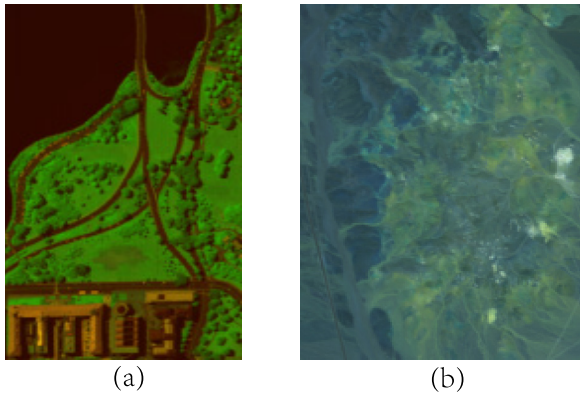


Fig. 8. (a) False color image of Washington DC Mall data (R: band 20, G: band 70, and B: band 150). (b) False color image of cuprite data (R: band 20, G: band 70, and B: band 150).

it shows the unmixing performance of six algorithms at different image sizes. By comparing the curves, we can conclude that as the image size increases, I-NLSU, CSNMF, $L_{1/2}$-NMF unmixing performance begins to increase, and then the performance declines. This is because as the size of the image increases, more and more spectral and spatial information can be utilized, and the performance of the algorithm will become better. However, as the image size is further increased, the unmixing performance gradually becomes stable. In addition, the output complexity of the algorithm increases as the image size increases. Therefore, it is also important to select the appropriate image size to ensure performance and efficiency during the unmixing process.

In this experiment, we also record the running time of each algorithm under different image sizes. All algorithms are performed on a Linux system with Intel (R) Xeon (R) CPU E5-2697 v3 at 2.60 GHz and 64-G RAM. The running time is shown in Table I. Obviously, our algorithm has an increase dramatically in run time as the image size increases. This is mainly because our algorithm uses both spatial and spectral

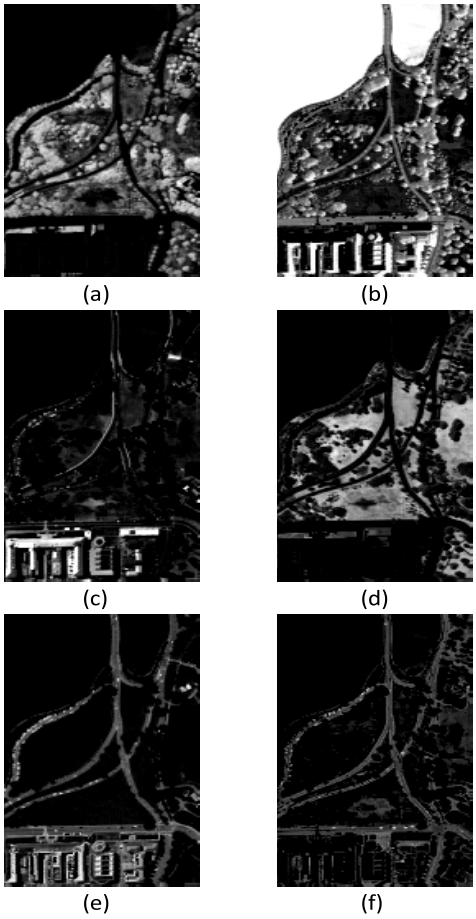


Fig. 9. Abundance maps of different endmembers using S^2-NMF on the Washington DC Mall data. (a) Tree. (b) Water/shadow. (c) Roof. (d) Grass. (e) Road. (f) Trail.

self-similarity constraints. When calculating two similarity matrices, the complexity of S^2-NMF will be larger. In addition, I-NLSU, CSNMF algorithms use spatial information, and their runtime also increases significantly as the image size increases, while NMF-QMV and SGSNMF is relatively stable. Through this experiment, we can understand that for hyperspectral unmixing, it is also an important problem to select the appropriate image size. The running time of the algorithm should also be considered on the basis of ensuring the performance of the algorithm.

B. Experiments on Real Hyperspectral Data

In this section, we use two real hyperspectral data to evaluate our proposed algorithm. The first data is Washington

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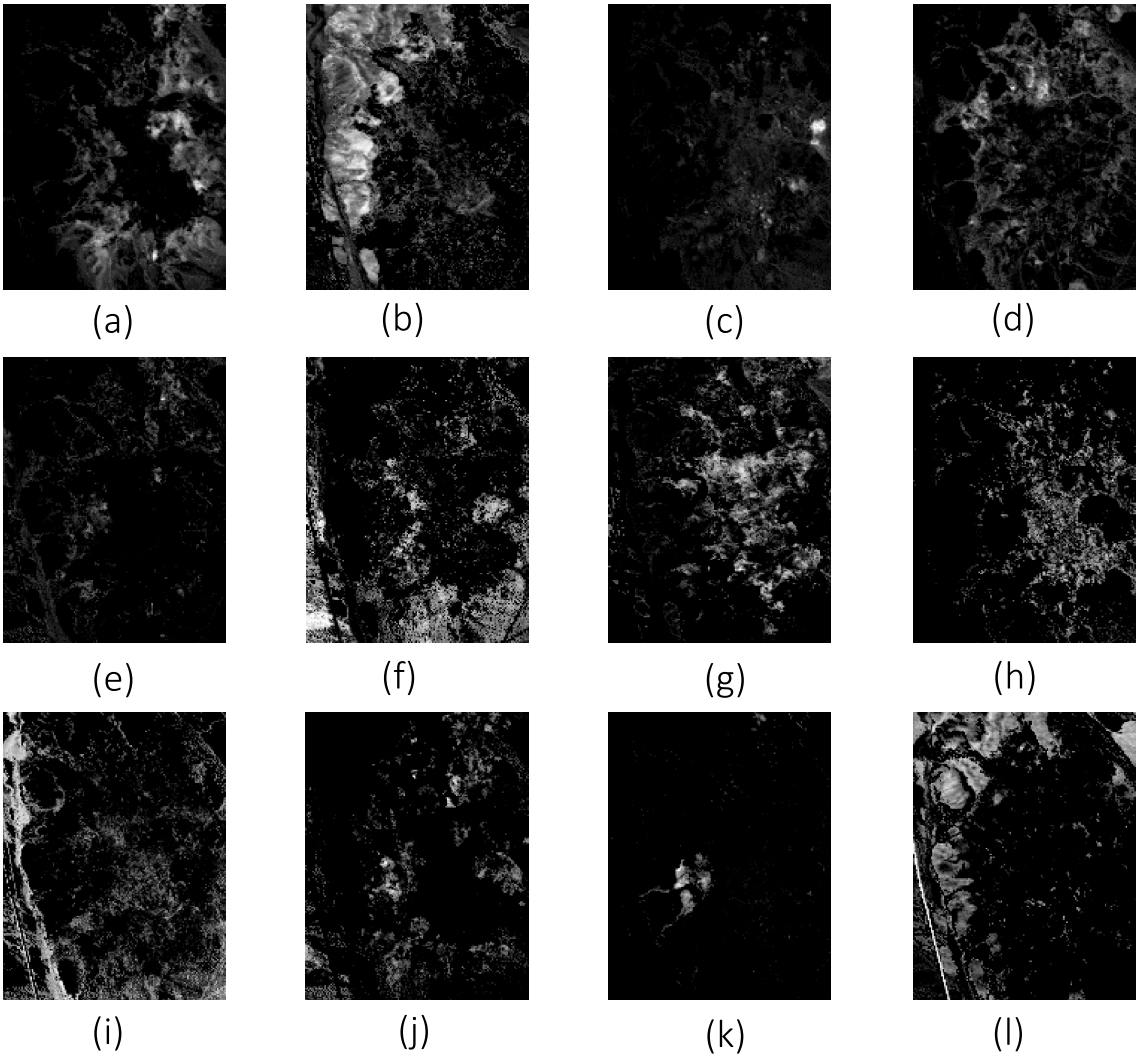


Fig. 10. Abundance maps of different endmembers using S^2-NMF on the AVIRIS cuprite data. (a) Alunite. (b) Sphene. (c) Muscovite. (d) Montmorillonite. (e) Kaolinite #2. (f) Dumortierite. (g) Chalcedony. (h) Nontronite. (i) Andradite. (j) Kaolinite #1. (k) Buddingtonite. (l) Pyrope.

TABLE II

SAD AND THE STANDARD DEVIATIONS (%) RESULTS ON THE WASHINGTON DC MALL DATA

	S^2-NMF	I-NLSU	$L_{1/2}$-NMF	CSNMF	NMF-QMV	SGSNMF
Roof	0.0427±1.23	0.1203±2.37	0.0815±1.52	0.1243±0.92	0.0531	**0.0229±0.61**
Water/Shadow	**0.0357±0.58**	0.0548±0.24	0.0520±0.29	0.2475±3.25	0.8680	0.0528±2.37
Grass	**0.0524±2.12**	0.1452±1.08	0.1886±7.68	0.0728±2.13	0.1157	0.1058±10.4
Tree	0.0728±2.06	0.0984±2.73	**0.0611±1.54**	0.0854±1.06	0.2924	0.1699±7.73
Road	0.1743±3.57	**0.1572±4.64**	0.1902±3.94	0.1687±4.32	0.1863	0.4588±15.4
Trail	**0.1602±3.15**	0.1847±2.98	0.3125±15.6	0.1842±3.26	0.3365	0.4073±17.9
Mean	**0.0930±2.12**	0.1218±2.34	0.1209±5.09	0.1472±2.49	0.4086	0.1015±4.54

DC Mall Data,[2] which is captured by hyperspectral digital imagery collection experiment (HYDICE) in August 1995. The Washington DC Mall Data has a total of 210 bands, including wavelength ranging from 0.4 to 2.4 μm. Since some of the data are heavily disturbed by moisture and noise, these bands are removed before unmixing. Finally, only 191 bands are reserved for the data, and a 200×120 subregion is taken as the image to be unmixed. Fig. 8(a) is the false color image.

According to the results presented in [28], we can conclude that Washington DC Mall Data has six types of endmemebers, including roof, water/shadow, grass, trees, road and trail. Since this database does not provide real endmembers, we select 20 nearly pure pixels for each endmember and use their average as the real reference endmember. Table II shows the SAD and standard deviation results of the proposed method and other five comparison algorithms on the Washington DC Mall data. As shown in Fig. 9, it is the abundance maps obtained by S^2-NMF. To sum up, because our algorithm takes into account the characteristics of the joint representation of the spatial–spectral information, it can make full use of the context information contained in the HSI than other algorithms. Therefore, the unmixing performance is more superior. Since the real endmembers are simulated, the comparison of the endmember curves before and after unmixing is not shown here.

[2][Online]. Available: https://engineering.purdue.edu/biehl/MultiSpec/hyperspectral.html

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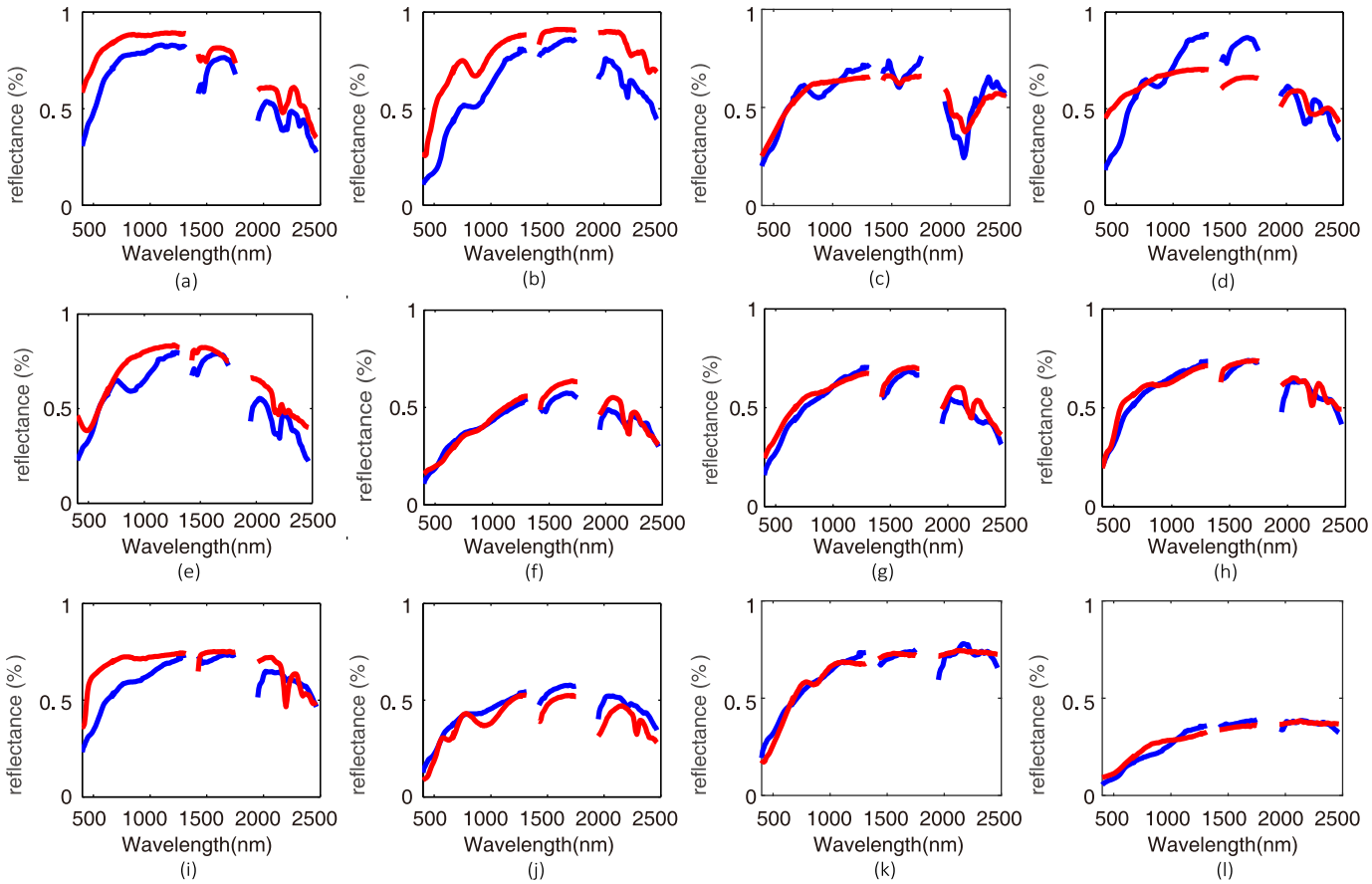


Fig. 11. Comparison of the USGS library spectra (red line) with the endmember signatures extracted by S^2-NMF (blue line) on the AVIRIS cuprite data. (a) Alunite. (b) Andradite. (c) Buddingtonite. (d) Chalcedony. (e) Dumortierite. (f) Kaolinite #1. (g) Kaolinite #2. (h) Montmorillonite. (i) Muscovite. (j) Nontronite. (k) Pyrope. (l) Sphene.

TABLE III
SAD AND THE STANDARD DEVIATIONS (%) RESULTS ON THE AVIRIS CUPRITE DATA

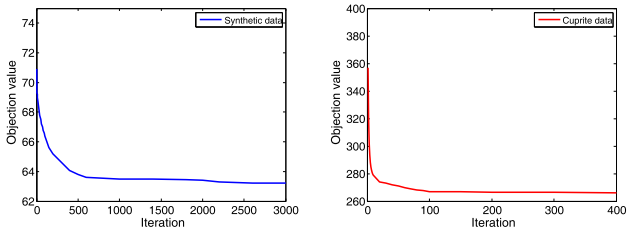
	S^2-NMF	I-NLSU	$L_{1/2}$-NMF	CSNMF	NMF-QMV	SGSNMF
Alunite	0.0956±2.32	0.1325±3.21	0.1660±0.61	0.1129±8.14	**0.0860**	0.1748±6.76
Andradite	**0.0650±1.49**	0.0694±1.83	0.0822±1.31	0.0886±2.76	0.1212	0.0688±0.21
Buddingtonite	**0.0635±0.31**	0.1072±2.09	0.0998±1.34	0.0900±1.81	0.0797	0.1094±1.62
Chalcedony	0.1102±2.80	**0.0824±0.85**	0.1307±7.34	0.1296±2.47	0.1428	0.1399±3.01
Dumortierite	0.1167±1.07	0.1036±1.92	0.0932±4.16	**0.0861±7.07**	0.1108	0.1044±1.00
Kaolinite#1	**0.0666±0.12**	0.0871±2.73	0.0778±8.43	0.0790±8.47	0.0795	0.1038±1.59
Kaolinite#2	**0.0594±0.67**	0.0649±1.03	0.0692±7.28	0.0704±5.24	0.1014	0.0878±2.72
Montmorillonite	**0.0477±0.35**	0.0683±2.66	0.0695±5.31	0.0637±3.16	0.1114	0.0811±0.99
Muscovite	0.1569±3.07	0.1673±2.48	0.1644±8.15	0.1443±9.24	**0.0811**	0.0841±2.81
Nontronite	0.0765±0.19	0.0895±1.57	0.0894±1.41	0.0861±0.65	**0.0715**	0.0844±1.05
Pyrope	0.0701±5.30	0.0632±3.14	0.0616±1.26	0.0861±3.06	0.1812	**0.1046±5.09**
Sphene	**0.0690±0.53**	0.0815±2.59	0.1628±10.5	0.1368±8.42	0.1879	0.1371±9.74
Mean	**0.0831±1.51**	0.0931±2.18	0.1056±4.75	0.0978±5.04	0.1190	0.1067±3.05

The second hyperspectral data is the cuprite captured by AVIRIS in 1997.[3] These data have a total of 224 bands. The range of wavelengths is from 0.37 to 2.48 μm. The region covered by the image contains a variety of minerals. In our experiment, the test image size is 200 × 150. Fig. 8(b) is the fake color image. Here is a similar operation to the Washington DC Mall Data unmixing preprocessing. After removing the bands that are affected by moisture and noise before unmixing, only 188 bands are remained. According to [6] and [7], this image has 14 endmembers. Since the difference in similar minerals is small, the number of endmembers is reduced to 12, including Alunite, Andradite, Buddingtonite, Chalcedony, Dumortierite, Kaolinite #1, Kaolinite #2, Montmorillonite, Muscovite, Nontronite, Pyrope, and Sphene.

The abundance map obtained by the proposed algorithm is shown in Fig. 10. As shown in Fig. 11, it is a comparison of the endmember spectra obtained by S^2-NMF unmixing and the real endmembers provided by the USGS library. As can be seen from Fig. 11, the estimated endmembers and the real endmembers are very close, which proves the effectiveness of the proposed algorithm. Table III shows the SAD of the six algorithms in this experiment and their standard deviations. We can conclude that our proposed algorithm not only provides the most accurate

[3][Online]. Available: http://aviris.jpl.nasa.gov/data/freedata.html

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(a) (b)

Fig. 12. Convergence analysis of S^2-NMF on (a) synthetic data and (b) cuprite data.

unmixing performance, but also performs well in algorithm stability.

C. Convergence Analysis

Here, we use two data to illustrate the convergence of the proposed method, one of which is synthetic data of image size 20×20 in Section V-A: experiment 5, and the other is cuprite data. The convergence process of S^2-NMF on two data is presented in Fig. 12, where we show how the objective function value Γ changes as the number of iterations increases. As shown in Fig. 12, the proposed algorithm converges well on both synthetic and real data. Since synthetic data uses a linear mixing method and is limited by noise, Γ is much smaller than the cuprite data and decreases less. S^2-NMF on synthetic data converges in about 600 iterations while cuprite data converge in about 200 iterations. Therefore, we set a maximum number of iterations of 3000 is reasonable in this article, and this experiment can verify that the proposed algorithm can converge.

VI. Conclusion

In this article, we propose a novel hyperspectral unmixing method called S^2-NMF, which combines spatial and spectral information of data. In order to obtain more robust results in the unmixing, we explore the global spatial structural group similarity and local spectral group similarity in the HSI. And then the abundance vectors of the pixels within each group are forced to be similar according to the self-expression characteristics of the data in the subspace. The experiments on synthetic data and real data prove the robustness of our proposed algorithm, and also confirm the role of our proposed self-similarity constraint in exploring the spectral information of hyperspectral data.

In addition, by analyzing the results of the unmixing of synthetic data and real hyperspectral data, it is clear that the results of synthetic data are better than real experiments. This is because real data not only has a lot of noise interference, but also is not a single linear combination. Therefore, in the future work, we will focus on exploring models that are more suitable for hyperspectral unmixing.

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